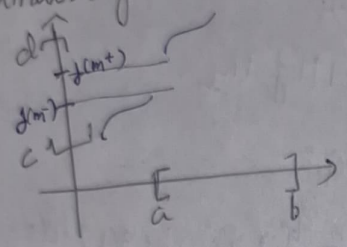
18/08

Lec-8

Result:→ If monotone, then the f has countably many jump discontinuity

Theorem:- If a function if $f:[a,b]\to[c,d]$ is monotone and onto then f is continuous function.



f is discontinuous at $a\in(a,b)$

$f(a^-) < f(a^+)$ (→ Strictly less than)

$f(a^-), f(a^+) \subset [c,d]$

there is jump discontinuity bec of montonicity

Corollary:- Centre Lebesgue jor is continuous.

Metric Spaces

Take a set of function.

$C[0,1] = \{f : [0,1]\to\mathbb{R} : f \text{ is continuous}\}$

$f_0 \in C[0,1]$.

"Metric is giving concept of distance in a general/abstract set"

Defination:- let M be a set. Consider a function $d: M\times M\to\mathbb{R}$ satisfying:

i) $d(x,y)\geq 0$ for any $x,y\in M$

ii) $d(x,y)=0$ if and only if $x=y$.

(distance = d function. Same points then d is 0.)

iii) (Symmetry) $d(x,y) = d(y,x)$

iv) (Triangle inequality) $\forall\ x,y,z\in M$

$d(x,y)\leq d(x,z) + d(z,y)$

Then we call d as a Metric and (M, d) as a Metric Space.

Note: Abstract Sets: $\mathbb{R}^2, \mathbb{R}^3, \mathbb{R}^n$ general sets which we can visualise.

Example:-

i) Euclidean metric on $\mathbb{R}$ $d, \mathbb{R}\times\mathbb{R}\to\mathbb{R}^+$

$d(x,y) = |x-y|$

ii) $(\mathbb{R}^2, d)$ is defined by $x=(x_1,x_2)$, $y=(y_1,y_2)$

$d(x,y) = \sqrt{(x_1-y_1)^2+(x_2-y_2)^2}$ → ? Check.

iii) Given a set M; $d: M\times M\to[0,\infty)$ by

$$d(x,y) = \begin{cases} 0 & \text{if } x=y \\ 1 & \text{if } x\neq y \end{cases}$$

Q→ Is (M,d) a metric?? yes.

this metric is called a discrete metric and (M, d) is called discrete Matric space.

Remark:- Any set M, admits atleast one metric.

① If $A\subseteq M$; and (M, d) is a metric space then (A, d) is also a metric space.

② If for $\mathbb{R}^n$, $n>1$;

$$d_1(x,y) = \left(\sum_{i=1}^{n} |x_i-y_i|^2\right)^{1/2}$$

$$d_2(x,y) = \sum_{i=1}^{n} |x_i-y_i|$$

$d_3(x,y) = \max_{1\le i\le n} [|x_i - y_i|]$

d_1, d_2, d_3 all are Metric spaces in $\mathbb{R}$

Normed vector Space

Let V be a vector space.

Defn:- $\|\cdot\| : V \to [0,\infty)$ is a norm. if it satisfies.

(i) $\|x\| \ge 0$ and $\|x\| = 0$ iff $x = 0$

ii) $\alpha \in \mathbb{R}/\mathbb{C}$. and $x \in V$ then $\|\alpha x\| = |\alpha| \|x\|$

iii) $\|x+y\| \le \|x\| + \|y\|$

Let $(V, \|\cdot\|)$ be a normed vector space, Then we define a metric by $d(x,y) = \|x-y\|$

Example:- Consider; $C[a,b]$;

Then (i) $\|f\|_1 = \int_a^b |f(t)|\,dt$ (L^1-norm)

(ii) $\|f\|_2 = \left(\int_a^b |f(t)|^2 dt\right)^{\frac{1}{2}}$ (L^2-norm)

iii) $\|f\|_\infty = \max_{t\in[a,b]} |f(t)|$ (L^∞-norm)

All 3 define a norm on $C[a,b]$.

Remark:

$L^p[a,b] = \{f : [a,b] \to \mathbb{R} : \|f\|_p = \left[\int_a^b |f(t)|^p\right]^{\frac{1}{p}} < \infty\}$

Example:- Take $g = \frac{1}{\sqrt{x}}$, then $x \in [0,1]$, $g(0) = 0$

$g \in L^1[(0,1)]$

as $\int_a^b \frac{1}{\sqrt{x}}\,dx$ is finite But $g \notin L^2[(0,1)]$

Check:- $\|\cdot\|_1$ is a norm on $C[a,b]$.

To check the norm defination.

i) is trivial.

To see ii) $\alpha \in \mathbb{R}$, $f \in C[a,b]$

$$\|\alpha f\|_1 = \int_a^b |\alpha f(t)|\,dt$$

$$= \int_a^b |\alpha||f(t)|\,dt$$

$$= |\alpha| \|f\|_1$$

To see iii)

$$\|f+g\|_1 = \int_a^b |f(t)+g(t)|\,dt$$

$$\le \int_a^b (|f(t)| + |g(t)|)\,dt$$

$$= \int_a^b |f(t)|\,dt + \int_a^b |g(t)|\,dt$$

$$= \|f\|_1 + \|g\|_1$$ proved

Example:- Set of $\{x_n\} \to$ seq where $x_n \in \mathbb{R}$; One can check that this is a vector space (Exercise)

(i) By $x = \{x_n\}$;

Take $\|x\|_1 = \sum_{n=1}^{\infty} |x_n|$

$\|x\|_p = \left(\sum_{n=1}^{\infty} |x_n|^2\right)^{\frac{1}{p}}$

$\|x\|_\infty = \sup x_n$

Now, we define

$l_p = \{\{x_n\} \in \mathbb{R} : \|x_n\|_p < \infty\}$ for $1 \le p < \infty$

$l_\infty = \{\{x_n\} \in \mathbb{R} : \|x_n\|_\infty < \infty\}$

Show: $(l_p, \|\cdot\|_p)$ is a normed vector space.

Cauchy - Schwarz Ineq.

$$\sum |x_i \cdot y_i| \leq \left(\sum |x_i|^2\right)^{\frac{1}{2}} \left(\sum |y_i|^2\right)^{\frac{1}{2}}$$

Example: Assume (M, d) is a Metric Space.

We define a new metric by:

$$d_1(x,y) = \min\{d(x,y), 1\}$$

Check (M, d_1) is a metric space.

Note $d_1(x,y) \geq 0$.

Now if $d_1(x,y) = 0 \Leftrightarrow d(x,y) = 0$

$$\Leftrightarrow x = y$$

If (Symmetry)

$$d_1(x,y) = d(x,y) = d(y,x) = d_1(y,x)$$

As $d_1(x,y) = d(x,y)$

$\Rightarrow d(x,y) \leq 1$

$d(y,x) \leq 1$.

Then $d_1(y,x) = d(y,x)$

Similarly if $d_1(x,y) = 1 = d_1(y,x)$

Triangle Inequality :- given $x, y, z \in M$

To Show, $d(x,y) \leq d_1(x,z) + d_1(y,z)$

Case1:-

Assume $d_1(x,z) \geq 1$ or $d_1(y,z) \geq 1$

Then $d_1(x,z) + d_1(y,z) \geq 1 \geq \min\{d(x,y), 1\}$

$$= d_1(x,y)$$

Case 2:

Both $d_1(x,z) < 1$ and $d(y,z) < 1$

Then $d_1(x,z) + d_1(y,z) = d(x,z) + d(y,z)$

$$\geq d(x,y)$$

$$\geq \min\{d(x,y), 1\}$$

$$= d_1(x,y)$$

Therefore: (M, d_1) is a Metric Space.

Open Balls in Metric Space.

Let (M, d) be a metric space

$$B_r(x) = \{y \in M : d(x,y) < r\}$$

Eg For $(\mathbb{R}, d_1)$ what is $B_2(0)$.

20/8/20

Rev-9

- (M,d) Metric space
  ↑ metric $d: M \times M \to [0, \infty)$

i) $d(x,y) \geq 0$

ii) $d(x,y) = 0 \iff x = y$

iii) $d(x,y) = d(y,x)$

iv) $d(x,y) \leq d(x,z) + d(y,z)$

Open Ball :-

$B_r(x) = \{ y \in M, d(x,y) < r \}$.

Eg $(\mathbb{R}^n, d_1)$ Metric Space

$d_1(x,y) = \min \{ \|x-y\|_2, 1 \}$

What about

$B_1^{d_1}(0) = B_1^{\|\cdot\|_2}(0) = \{ y : \|y\|_2 < 1 \}$

$B_2^{d_1}(0) = \mathbb{R}^n$.

Example :-

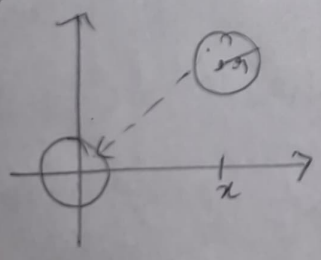
- (M,d) be a discrete metric.

Then $B_1^d(x) = \{x\} = B_{\frac{1}{2}}^d(x)$ for $r \leq 1$

$B_2^d(x) = M - B_R^d(x)$ for $R \geq 1$

(*) $(V, \|\cdot\|)$ normed vector space

$B_r(x) = x + B + B_r(0) = \{ x + y : y \in B_r(0) \}$



$B_r(x) = x + B_r(0) = x + r B_1(0)$

$= \{ x + ry : y \in B_1(0) \}$

Bounded Set :- $(\mathbb{R}, d)$ A is bounded if and

$|x| < m \quad \forall x \in A$

$A \subseteq (-m, m)$

(M,d) distance → metric space

A set $A \subseteq M$

is bounded if $A \subseteq B_r(x)$

for some $r > 0$, $x \in M$.

And r and x are not unique.

Example :- For $(\mathbb{R}^n, d_1)$ $[d_1 = \min\{\text{norm}, 1\}]$

$\mathbb{R}^n$ is a bounded set.

↳ (depending on the metric)

Defination :- Diameter

$\text{diam}(A) = \sup \{ d(a,b) : a, b \in A \}$ dist function

Defination :- neighbourhood of a point.

A nbd of a point x is any set

containing an open ball centered at x.

For example, $x \in A$, if $\exists r > 0$ s.t

$B_r(x) \subseteq A$

then A is a nbd of x.

Convergent Seq :- (M,d) → Metric space

we say $\{x_n\} \subset M$ is converging to $x \in M$

for any given $\varepsilon > 0$, $\exists N_\varepsilon \in \mathbb{N}$ s.t

$d(x_n, x) < \varepsilon, \quad \forall n \geq N_\varepsilon$

Cauchy Seq :- A seq is cauchy if for any

$\varepsilon > 0$ $\exists N_\varepsilon \in \mathbb{N}$ s.t

$d(x_n, x_m) < \varepsilon \quad \forall n, m \geq N_\varepsilon$

Example :- $M = [0,1]$.

$x_n = \frac{1}{n} \to$ is not converging to any point in M.

(1) $(\mathbb{R}, d) \to$ discrete Metric Space.

$x_n = n$;

Is x_n Convergent?

$\{x_n\}$ is a bounded seq but it does not have any Convergent Subseq

(2) For discrete metric Space A Cauchy seq is always a constant seq (after some finite number).

If $\{x_n\}$ is Cauchy $\iff x_n = a \ \forall n \geq N, \ a \in M$

Equivalently;

For any $\varepsilon > 0$, we should have $\{x_n : n \geq N_\varepsilon\} \subseteq B_\varepsilon(x)$

[We say a seq $\{x_n\}$ is eventually in 'A' if $\exists N \in \mathbb{N}$ s.t. $\{x_n : n \geq N\} \subset A$]

Equivalently;

$\{x_n\}$ converges to x if $\{x_n\}$ is eventually in every nbd of x.

Remark :- $(\mathbb{R}^d, \|\cdot\|)$ → metric Space

$\|\cdot\|_1, \|\cdot\|_2, \|\cdot\|_\infty$.

$\{x_n\} \subset \mathbb{R}^d$ Now if $x_n \to x$ Then we know that it also Converges coordinate wise

$\{x_n^1, x_n^2, \ldots, x_n^d\}$

Indeed, $x_n^i \to x^i \ \forall i \in \{1, \ldots, d\}$

$\|x_n\| \geq |x_n^i|$

Converse, is also True !!! (Why ?)

But, this is not the case for $l_1, l_2, l_3, \ldots, l_\infty$ although the norms are generalisation of norms in $\mathbb{R}^d$.

$l_1 = \{x_n\} : \sum_{n=1}^{\infty} |x_n| < \infty\}$

If $\{x_n\} \subset l_1$ converges then $x_n = (x_n^1, x_n^2, x_n^3, \ldots)$ x_n^i also converges !!

as $\sum_{i=1}^{\infty} |x_n^i| \geq |x_n^i|$

But, co-ordinate wise convergence does not imply the seq convergence in $l_1, l_2, l_3 \ldots l_\infty$

Proof Given $x_n \to x$ in $(\mathbb{R}^d, \|\cdot\|_1)$

given $\varepsilon > 0$ $\exists N_\varepsilon$ s.t.

$|x_n^i - x^i| \leq \|x_n - x\|_1 < \varepsilon \ \forall n > N_\varepsilon$

$e_k = (0, 0, \ldots, 1, 0, \ldots) = (e_k^1, e_k^2, \ldots)$ (1 at k^{th} place)

$e_k \in l_1 \to e_k^i \to 0$ as $k \to \infty$

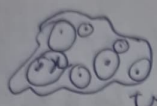
But $\to (0, 0, \ldots)$ a seq

$\|e_k - 0\|_1 = 1 \ \forall k \in \mathbb{N}$

Even $\|e_k - e_m\| \geq 1$ if $k \neq m$

Open Set & Closed set

Defination :- Let (M, d) be a Metric Space. A set $U \subseteq M$ is called an open set if for every $x \in U$, $\exists \varepsilon_x > 0$ such that $B_{\varepsilon_x}(x) \subseteq$



↳ is it open or not? We are seeing.

Similarly A set $F \subseteq M$ is called a closed set if F^c is open.

$F^c = M \setminus F$

Properties:

i) M is an open set; Also the null set ϕ is an open set (by default)

ii) $([0,1], d)$ is a Metric space.

then $[0,1]$ is an open set in (M,d)

iii) M is also a closed set, ϕ is also closed set.

iv) open balls for $([0,1],d)$.

$[0,a)$, $a<1$

$[b,1]$ or $b<1$

(c,d) $c,d \in [0,1]$

$[0,a) = (-1,a) \cap [0,1]$ (written as)

$\Downarrow$

open Ball in $(\mathbb{R},d)$

28/Aug/2024 ⇒ Monday

Lec-10

Open set :-

A subset of (M, d)

A is open : $x \in A \Rightarrow \exists \varepsilon > 0$ s.t $B_\varepsilon(x) \subset A$ is closed

$M \setminus A = A^c$ is open set

Example :-

i) $(\mathbb{R}, d) \to (a, b) \to$ open set

$[a, b] \to$ closed set

$[a, b) \to$ Not open also not a closed set.

ii) $(\mathbb{R}, \text{discrete})$

$\Rightarrow \{x\} \to$ open set / closed set.

Theorem :- Given any set A in (M, d) the following are equivalent :

i) A is closed i.e A^c open

ii) If $B_\varepsilon(x) \cap A \neq \phi \ \forall \varepsilon > 0 \Rightarrow x \in A$

iii) If a seq $\{x_n\} \subset A$ converges to $x \in M$; then $x \in A$.

Proof :- i) $\Rightarrow$ ii)

Given A is closed $\Rightarrow A^c$ is open

Let $x \in A^c$. $\exists \varepsilon_0 > 0$ s.t

$B_{\varepsilon_0}(x) \subset A^c$

$B_{\varepsilon_0}(x) \cap A = \phi$

$\Rightarrow$ If $B_{\varepsilon_0}(x) \cap A \neq \phi \ \forall \varepsilon > 0$.

implies $\Rightarrow x \in A$

ii) implies $\Rightarrow$ i) :-

Given (ii) :

If $x \in A^c$, $\exists \varepsilon_0 > 0$ s.t

$B_{\varepsilon_0}(x) \cap A = \phi$

implies $\Rightarrow B_{\varepsilon_0}(x) \subseteq A^c$

A^c is open or A is closed.

ii) $\Rightarrow$ (iii) :-

Let $\{x_n\} \subset A$, and $x_n \to x \in M$

Claim $x \in A$

for any $\varepsilon > 0$ $d(x_n, x) < \varepsilon \ \forall n \geq N_0$

$\Rightarrow B_\varepsilon(x) \cap A \neq \phi$

$\Rightarrow x \in A$ (from ii statement)

iii) $\Rightarrow$ ii)

$x \in M$ if $\forall \varepsilon > 0$.

$B_\varepsilon(x) \cap A \neq \phi$.

$\exists \ x_n \in A$ such that

$x_n \in B_{\frac{1}{n}}(x) \cap A$

$\Rightarrow$ we get $\{x_n\} \subset A$

s.t $x_n \to x$ $x \in A$.

Example :- $(\mathbb{R}, d)$; $A \to$ finite set

$d \to$ usual metric

$\Rightarrow A$ is closed.

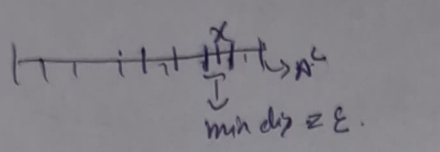
$\Rightarrow A = \{a_1, \dots, a_n\}$

$\varepsilon = \frac{1}{2} \min [a_i - a_j] \to$ distance $i \neq j$

~~$x \notin A$ $B_\varepsilon(x) \cap A = \phi$~~

Let $x \in A^c$.

$\varepsilon = \frac{1}{2} \min\{|x - a_i| : a_i \in A\}$



Then take neighbourhood which will close to x and open

$B_\varepsilon(x) \cap A = \phi$.

By ii) ~~point~~

$x_n \in \{a_1, \dots, a_n\}$

and $x_n \to x \in \mathbb{R}$. after some number.

$x_n = a_i$ for some a_i

Interior of a set

Definition:-

Given a set A in (M,d) metric space. we define the interior of A by.

$Int(A) = A^\circ$

$= \cup\{V : V \subseteq A \text{ and } V \text{ is open}\}$

$= \cup\{B_\varepsilon(x) : B_\varepsilon(x) \subseteq A \text{ for some } x \in A,\ \varepsilon > 0\}$

$= \{x \in A : B_\varepsilon(x)) \subset A \text{ for some } \varepsilon > 0\}$

Example:-

i) $[a,b)$ $x \in (a,b)$

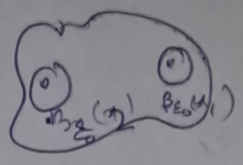
$x \in [a,b)^\circ$ as $\notin [a,b]^\circ$

ii) $(0,1)^\circ = (0,1)$

To show $A_1 = A_2$

$A_1 \subseteq A_2$ and $A_2 \subseteq A_1$

$x \in \cup\{U : U \subseteq A \text{ open}\}$



$\Rightarrow x \in U_\alpha \subset A$

open set

$\Rightarrow \exists\ \delta_0 \ \forall\ t$

$B_\varepsilon(x) \subset U_t \subset A$

$\Rightarrow x \in A_2$

Property:- A° is the biggest open set Contained in A.

Closure of A

given (M,d) metric space.

$A \subset M$ we define:- Closure of A.. $Cls(A) = \bar{A}$.

$= \cap\{F : A \subseteq F \text{ and } F \text{ is closed}\}$

Result:- $\bar{A}$ is a closed set and $\bar{A}$ is the smallest closed set Containing A.

Characterization

Lemma:- $x \in \bar{A}$ iff and only if

$B_\varepsilon(x) \cap A \neq \phi \ \forall \varepsilon > 0$

Corollary:- $x \in \bar{A}$ iff $\exists \{x_n\} \subseteq A$ s.t $x_n \to x$

Thursday

Recall: $f:(M,d) \to (N,\rho)$

for any $\varepsilon > 0$ continuous

$\exists\, \delta > 0$ $\quad B_\delta^d(a) \subseteq f^{-1}(B_\varepsilon^\rho(f(x)))$

Theorem $f: M \to N$, then the following are equiv.

i) f is continuous on M

Continuous at each.

ii) If $E \subset N$ is a closed set in N, then $f^{-1}(E)$ is also a closed set.

iii) If $V \subset N$ is open then $f^{-1}(V)$ is open in M.

Proof: (i) $\Rightarrow$ ii)

given f is continuous.

Let $\{x_n\} \subset M$ converging seq st

$x_n \xrightarrow{d} x_0$

given any $\varepsilon > 0$, $\exists \delta > 0$ st

$B_\delta^d(x) \subseteq f^{-1}[B_\varepsilon^\rho(f(x_0))]$

as $x_n \xrightarrow{d} x_0 \Rightarrow x_n \in B_\delta^d(x_0)$ $\forall n \geq n_\delta$

$\Leftrightarrow d(x_n, x) < \delta$

Hence $x_n \in f^{-1}(B_\varepsilon^\rho(f(x)))$

$\Rightarrow f(x_n) \in B_\varepsilon^\rho(f(x))$ $\forall n \geq N_\delta$

$\Leftrightarrow \rho(f(x_n), f(x)) < \varepsilon$ $\forall n \geq$

ii) $\Rightarrow$ iii)

Given E is a closed set in N

Aim to Show $f^{-1}(E)$ is closed

$f^{-1}(E) \subseteq M$

Take a converging seq.

$\{x_n\} \subseteq f^{-1}(E)$

st $x_n \to x$

we note $f(x_n) \in E$

By (ii) we know

$f(x_n) \to f(x)$

given E is closed

$f(x) \in E$

$\Rightarrow x \in f^{-1}(E)$

Hence $f^{-1}(E)$ is closed

iii) $\Rightarrow$ iv)

Check $f^{-1}(A^c) = (f^{-1}(A))^c$

prove the part.

iv) $\Rightarrow$ i)

Assume (iv) is true

given any $\varepsilon > 0$ $B_\varepsilon^\rho(f(x))$ $x \in M$ is an open set in N.

Then by (iv) $f^{-1}(B_\varepsilon(f(x)))$ is open

As $x \in f^{-1}(B_\varepsilon f(x))$ open set

$\exists\ \delta > 0$ s.t $B_\delta(x) \subseteq f^{-1}(B_\varepsilon(f(x)))$

$\therefore f$ is continuous.

Examples :-

$f:(\mathbb{N}, d) \to (\mathbb{R}, d)$

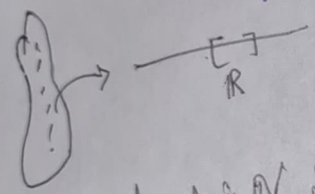
$d \to$ usual metric

$d(x,y) = |x-y|$

f is a function.

To see if f is continuous Take a ~~open~~ ~~closed~~ set $V \subseteq \mathbb{R}$. $f^{-1}(V) \to$ open

f is continuous.



Any closed set in $\mathbb{N}$ say A

$A = \mathbb{N} \cap F$

F is a closed set in $\mathbb{R}$

$\{1\} = \mathbb{N} \cap [0.9, 1.1]$

$\{2\} = \mathbb{N} \cap [1.9, 2.1]$

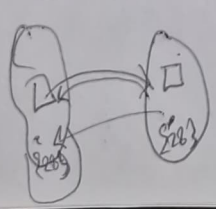
So, $\{2\}$ is both open and closed.

Eg :-

$i : (\mathbb{R}, d) \to (\mathbb{R}, d)$, $d = |x-y|$

s.t $i(x) = x$.

$i \to$ is continuous or not



identity map

i is continuous

$\tilde{f} : (\mathbb{R}, d) \to (\mathbb{R}, d')$

$d' = \begin{cases} 1 & x \neq y \\ 0 & x = y \end{cases}$

~~Homeo~~.

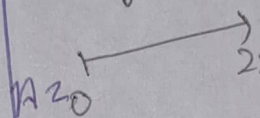
Homeomorphism

Defination :- We say two metric are homeomorphic if there is a map $f:(M,d) \to (N,F)$ s.t

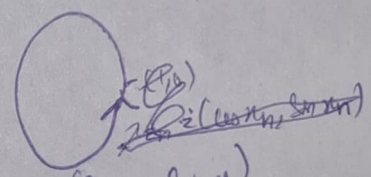
i) f is bijective

ii) f is continuous

iii) f^{-1} is continuous



$S^1 =$



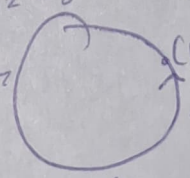
$f(x) = (\cos x, \sin x)$

$f : A \to S^1$ (bijective map)

f^{-1} is not continuous.

Also, $A = [0, 2\pi)$

$S^1 =$



given $z_n \xrightarrow{\text{converges}} (1,0)$

$f^{-1}(z_n) = x_n \longrightarrow 2\pi$

As $f^{-1}(1,0) = 0$

$f^{-1}(z_n) \to 0$ not possible

Def^n – Complete Metric Space – A metric space (M,d) is complete if every cauchy seq^n in M converges to a pt. in M.

Thm – Let (M,d) be a metric space & let $A \subseteq M$, then (A,d) is complete iff A is closed.

Proof – Exercise

10/9/25

Compactness – A metric space (M,d) is called compact if M is complete & totally bounded.

Eg: (i) $(\mathbb{R},d)$ → usual metric
$\mathbb{R}$ is not compact (not totally bdd).

(ii) $[a,b]$ → compact

(iii) (a,b) → not complete (not every cauchy seq^n will converge to a pt in (a,b) Take $a+\frac{1}{n}$)

Result – In $(\mathbb{R}^n,d)$, a set is compact iff its closed & bdd.

☆ $(l_1, \|\cdot\|_1)$ → closed bdd sets may not be compact.

$A = \{e^{(n)} : n \in \mathbb{N}\}$

$e^{(n)} = (0,0,\dots,\underset{\hookrightarrow n\text{th place}}{1},0,0,\dots)$

A is closed & bdd but not totally bdd.

Thm – (Compactness ⇔ sequential compactness) – (M,d) is a compact metric iff every seq in M has a convergent subseq.

Proof – M is compact ⇔ M is totally bdd + complete
⇒ Any seq^n in M has a "cauchy subseq" & by completeness cauchy subseq r convergent. □.

Corollary – (M,d) is a metric space. Let A be a subset of M. If A is compact, then A is closed. Also if M is compact and $A \subset M$ is closed, then A is compact.
So a closed subset of a compact metric space is compact.

Proof – 1st part
Take $\{x_n\} \subseteq A$ s.t. $x_n \to x$. Any subseq $\{x_{n_k}\} \subseteq \{x_n\}$ which is converging to $x \Rightarrow x \in A$. Hence, A is ~~con~~ closed.
2nd part – given A is closed. Let $\{x_n\} \subset A$ be any seq. As M is compact, $\exists$ $\{x_{n_k}\} \subseteq \{x_n\}$ s.t $x_{n_k} \to a \in M$.
As A is closed, $a \in A$. So for any seq $\{x_n\}$, we find a subseq^{n} $\{x_{n_k}\}$ converging to $a \in A$. Hence, A is compact.

Thm: Let $f: (M,d) \to (N,\rho)$ be continuous. If $K \subset M$ is compact ~~then~~ then $f(K)$ is compact.
Proof – Let $\{y_n\} \subseteq f(K)$. $y_n = f(x_n)$ $\forall \{x_n\} \subset K$
As K is compact, $\exists$ $\{x_{n_k}\} \subseteq \{x_n\}$ s.t. $x_{n_k} \to x \in K$
As f is cont., $f(x_{n_k}) \to f(x)$ $f(x)$ Hence $\{y_n\}$ has a convergent subseq in $f(K)$, hence $f(K)$ is compact □.

Corollary – Let (M,d) be compact. If $f: M \to \mathbb{R}$ is cont, then f is bounded & f attains both maxima & minima. ie. $\exists$ $x_0, y_0 \in M$ s.t. $f(x_0) = \max_{x \in M} f(x)$

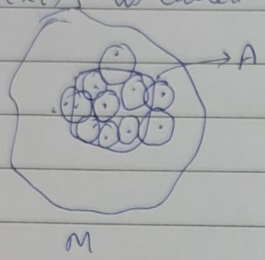
$$f(y_0) = \min_{x \in M} f(x).$$

Corollary – If M is compact then $\|f\|_\infty = \max_{x \in M} |f(x)|$ is a norm in $C(M)$ where $C(M) = \{f: M \to \mathbb{R} : f \text{ is cont}\}$

☆ Any f^n $f: M \to \mathbb{R}$ under discrete metric is continuous

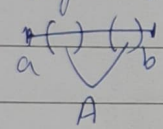
Totally Bounded Set - A set $A \subseteq M$ (where (M,d) is a metric space is called totally bdd if for any given $\epsilon > 0$ there are finitely many pts $x_1, \dots, x_{N_\epsilon}$ s.t. $A \subseteq \bigcup_{i=1}^{N_\epsilon} B_\epsilon(x_i)$

☆ $\{B_\epsilon(x_i)\}$ is called covers of A.



→ So, if we get a finite no. of pts in M s.t. the union of ball around em all of same radius will cover the whole A.

Eg: (i) $(\mathbb{R}, d)$ any bdd set is totally bdd.
Any bdd ~~interval~~ set is within an interval whose length is always finite.



$x_n = a + \frac{n\epsilon}{2}$ then use archimedian property

(ii) Discrete Metric space - (M,d)
$A = \{a_1, \dots a_n\}$ → finite → totally bdd ✓

For an infinite set, $A = \bigcup_{x \in A} B_\epsilon(x_\xi) = \bigcup_{x \in A} \{x\}$ $\forall \epsilon < 1$
∵ A is " $\Rightarrow \bigcup_{x \in A} \{x\}$ → infinite ⇒ not totally bdd

☆ For $\mathbb{R}$, a set is bdd $\iff$ totally bdd.

Thm - A set is totally bdd iff given any $\epsilon > 0$ there are finitely many open set $A_1, A_2, \dots, A_n \subset A$ st ∀ of $\text{diam}(A_i) < \epsilon$ $\forall i$ s.t. $A \subseteq \bigcup_{i=1}^{n} A_i$

Proof: Suppose A is totally bdd.
for any $\epsilon > 0$, $\exists n_\epsilon$ s.t. $A \subset \bigcup_{i=1}^{n} B_\epsilon(x_i)$ where $x_i \in M$ $\forall i \in \{1, \dots n_\epsilon\}$

Take $A_i = A \cap B_{\frac{\epsilon}{2}}(x_i)$

$\operatorname{diam}(A_i) \leq \epsilon$ & $A \subseteq A \bigcup_{i=1}^{n} A_i$

Converse - Given $\epsilon > 0$, we know $A \subset \bigcup_{i=1}^{n} A_i$ where $A_i \subseteq A \ \forall i$ $= \{1, \dots, n_\epsilon\}$ wd $\operatorname{diam}(A_i) < \epsilon$.

Given any $x_i \in A_i$, $A_i \subset B_{2\epsilon}(x_i)$

$A \subseteq \bigcup_{i=1}^{n_\epsilon} A_i \subseteq \bigcup_{i=1}^{n_\epsilon} B_{2\epsilon}(x_i)$. Then A is totally bdd. □.

$\Rightarrow$ Consider (l_1, d_1)

$l_1 = \{\{x_n\} : n \in \mathbb{N}, \text{ and } \sum_{i=1}^{\infty} |x_i| < \infty\}$ & $d = \sum_{i=1}^{\infty} |x_i|$ $\left(l_1 = \{(x_1, x_2 \cdots) : n \in \mathbb{N} \ \& \ \sum_{i=1}^{\infty} |x_i| < \infty\}\right)$ (infinite)

$e^{(n)} = (0, 0, \cdots, \underset{n\text{th space}}{1}, 0, 0, \cdots) \Rightarrow e^{(n)} \in l_1$ & $\|e^{(n)}\|_1 = 1$

Consider $\|e^{(n)} - e^{(m)}\|_1 = 2$ $m \neq n$ $(e^{(n)} - e^{(m)} = (0, 0 \cdots, \underset{n\text{th}}{1} \cdots 0, -1, \cdots)$

$A = \{e^{(n)} : n \in \mathbb{N}\}$ $\longrightarrow$ bdd set in (l_1, d_1) $\rightarrow$ Recall def^n of bdd set in metric space (A set is bdd iff $\|x\| \leq M \ \forall x \in A$ for some $M > 0$)

Take $\epsilon = \frac{1}{2}$, we can't write A as union of finite # subsets of A - Check:-

So, A is not totally bdd (and A is bdd).

So, till now, there is no corelation b/w bdd & totally bdd

Lemma :- Let $\{x_n\}$ be a seq^n in (M, d) & let $A = \{x_n : n \in \mathbb{N}\}$ be the range.

(i) If $\{x_n\}$ is cauchy, then A is totally bdd.

(ii) If A is totally bdd then $\{x_n\}$ has a cauchy subsequence

Proof: (i) Given any $\epsilon > 0$, $\exists\ N_\epsilon \in \mathbb{N}$ s.t. $d(x_n, x_m) < \epsilon\ \forall\ n, m \geq N_\epsilon$

Take $\{x_n\}$ $\{x_n : n \geq N_\epsilon\}$ & diam $\leq 2\epsilon$

$A = \{x_1\} \cup \{x_2\} \cup \cdots \cup \{x_{N_\epsilon - 1}\} \cup \{x_n, n \geq N_\epsilon\}$

Now diam of all those subsets above are less than 2ϵ

Hence A is totally bounded.

→ diam of $\{x_1\} = 0 = \{x_2\} \cdots$

(ii) Assume A is totally bdd & A is infinite. Given any $\epsilon > 0$,

$A \subseteq \bigcup_{i=1}^{n} A_i$ s.t. diam $(A_i) < \epsilon$

Choose $\epsilon = 1$, $A = A \subseteq \bigcup_{i=1}^{A_{n_1}} A_i^1$ s.t. diam $(A_i^1) < 1$

As A is infinite at least one of A_i^1 are infinite.

As one of the A_i^1 has to be infinite, we write this set as A_1 & take an element $x_1 \in A_1$

Note A_1 is totally bdd.

$\epsilon = \frac{1}{2}$, as A_1 is totally bdd

$$A_1 \subseteq \bigcup_{i=1}^{n_2} A_i^2 \quad \& \text{ diam } (A_i^2) < \frac{1}{2}$$

Again as A_1 is ∞, $\exists$ a A_i^2 having infinite elements, say A_2 & take $x_2 \in A_2 \setminus \{x_1\}$. Inductively $\exists\ A_n$ s.t. diam $(A_n) \leq \frac{1}{n}$ consider $x_n \in A_n \setminus \{x_1, \cdots, x_{n-1}\}$ & A_n is ∞. Then the seq^n $\{x_n\}$ is cauchy

Note - $A_1 > A_2 > A_3 \cdots$ & diam $(A_k) < \frac{1}{k}$ □.

Thm - A set A is totally bdd iff every seq^n in A has a cauchy subsequence in A.

Proof: ($\Leftarrow$) Assume A is not totally bdd. There is some $\epsilon_0 > 0$ s.t. A can't be covered by finitely many ϵ-balls. By induction, we can find x_n s.t. $d(x_n, x_m) > \epsilon_0$ (Exercise).

Then $\{x_n\}$ is not cauchy ⇒⇐.

Date
Page

Assignment 4

5. $f:(0,1)\to\mathbb{R}$, $f(x)=\tan\left(\pi\left(x-\frac{1}{2}\right)\right)$

① Check if f is bijective

tan f^n is strictly ↑ing on $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

Also, the $x\to\pi\left(x-\frac{1}{2}\right)$ is also strictly ↑ing

$\Rightarrow$ The composition of both strictly monotone f^n is strictly monotone $\Rightarrow$ f is injective (one-one)

as $x\to 0^+$, $f(x)\to-\infty$

$x\to 1^-$, $f(x)\to+\infty$

$\therefore$ f is cont. & strictly ↑ing, its image is all of $\mathbb{R}\Rightarrow f$ is onto.

So, f is bijective

② Check continuity –

$x\to\pi\left(x-\frac{1}{2}\right)$ is cont.

tan is cont. on $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

Since the composition of cont. f^n is cont. $\Rightarrow f$ is cont.

③ Inverse continuity

$\because x=\frac{1}{2}+\frac{1}{\pi}\tan^{-1}(y)$

Define $f^{-1}:\mathbb{R}\to(0,1)$, $f^{-1}(y)=\frac{1}{2}+\frac{1}{\pi}\tan^{-1}(y)$

$\because \tan^{-1}(y)$ is cont. on $\mathbb{R}\Rightarrow f^{-1}(y)$ is cont.

So, f is a homeomorphism b/w $(0,1)$ & $\mathbb{R}$

Hence $(0,1)$ & $\mathbb{R}$ r homeomorphic.

one way ans.

8. Suppose B is dis connected.

Then $\exists$ open sets B_1 & $B_2\subseteq B$ s.t. $B_1\cup B_2=B$ & $B_1\cap B_2=\emptyset$

$A\subseteq B=B_1\cup B_2\Rightarrow A\subseteq B_1$ or $A\subseteq B_2$ (use result obtained in Q 7).

WLOG, let $A\subseteq B_1$. Let $x\in\bar{A}\cap V\Rightarrow\because x\in\bar{B_2}\Rightarrow$ (open set) $\exists\epsilon_0>0$ s.t. $B_{\epsilon_0}(x)\subseteq B_2$ ①

take $x\in B_2$ s.t. $\because B_2\subseteq B\subseteq\bar{A}\Rightarrow x\in\bar{A}$ $x\in\bar{A}\Rightarrow\forall\epsilon>0$, $B_\epsilon(x)\cap A\neq\emptyset$

$\Rightarrow x\notin B_1$ $(B_2\cap B_1=\emptyset)\Rightarrow x\notin A\Rightarrow\Leftarrow$ $\Rightarrow\forall\epsilon>0, B_\epsilon(x)\cap B_1\neq\emptyset$

take $\epsilon=\epsilon_0\Rightarrow B_{\epsilon_0}(x)\cap B_1\neq\emptyset$ ②

① $\Rightarrow\Leftarrow$ ②

So, B is connected

9. A cont. map that's not open :-

$f(x)=0$, $f:\mathbb{R}\to\mathbb{R}$.

- f is cont (const f^n's r always cont.
- f is not open – Take $U\subseteq\mathbb{R}$ to be open then $f(U)=\{0\}$. But $\{0\}$ is

→ birds ans

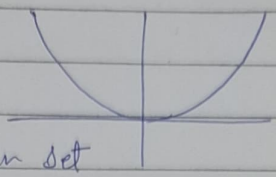
9 Cont. but not open –

$f: \mathbb{R} \to \mathbb{R}$ & usual metric

$f(x) = x^2$

$f(\mathbb{R}) = [0, \infty)$ → not an open set

↳ open set



Open but not cont –

$i: (\mathbb{R}, d) \to (\mathbb{R}, d_1)$

↳ usual ↳ discrete

Every subset of $(\mathbb{R}, d)$ is open so, its always an open ~~set~~ But i is not cont. (~~doesn't~~ every ~~does~~ subset in $(\mathbb{R}, d_1)$ is o but may not map to open image).

not open in ~~$\mathbb{R}$~~ $\mathbb{R}$.

Another eg: $g(x) = x^2 \to$ cont. but not open coz $g((-1,1)) = [0,1) \to$ not open

<u>An open map that's not continuous:-</u>

Take codomain $\mathbb{R}$ of discrete metric space $\rho(x,y) = \begin{cases} 1 & x \neq y \\ 0 & x = y \end{cases}$

The discrete metric space induces ~~to~~ the discrete topology where every subset is open & closed.

Let $f: (\mathbb{R}, d) \to (\mathbb{R}, d_1)$, $f(x) = x$

(d: usual in $(\mathbb{R}, d)$; d_1: discrete)

- If U is open, then $f(U) = U$ is also open in $(\mathbb{R}, d_1)$
- To be cont., the pre-image of ~~or~~ every open set in $(\mathbb{R}, d_1)$ must be open in ~~the~~ $(\mathbb{R}, d)$. But $\{x\}$ are open in the discrete codomain & $f^{-1}(\{x\}) = \{x\}$ which is not open in $(\mathbb{R}, d) \Rightarrow f$ is not cont.

3. Given any $\epsilon > 0$ if $d(x,y) < \delta$ for some $\delta \Rightarrow |f(x) - f(y)| < \epsilon$

$\iff |d(x,A) - d(y,A)| < \epsilon$

Aim:- $|d(x,A) - d(y,A)| \leq d(x,y)$ then we can take $\epsilon = \delta$.

Let $a \in A$. By Δ inequality, $d(x,a) \leq d(x,y) + d(y,a)$ $\forall a \in A$

$\inf_{a \in A} d(x,a) \leq \inf_{a \in A} \left(d(x,y) + d(y,a) \right) = \left(\inf_{a \in A} d(y,a) \right) + d(x,y)$

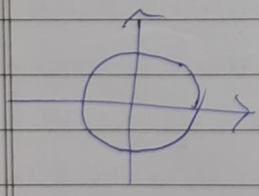
$\Rightarrow d(x,A) - d(y,A) \leq d(x,y)$

Taking $y = x$ & $x = y$, $d(y,A) - d(x,A) \leq d(x,y)$

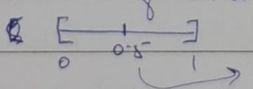
$\Rightarrow |d(x,A) - d(y,A)| \leq d(x,y)$. $\therefore$ □.

So it's Lipschitz cont $\Rightarrow$ cont.

$|f(x) - f(y)| \leq d(x,y)$

4. 

Assume $f: [0,1] \to S^1$ is a homeomorphism.



$\to$ unique map (as f is bijective)

Consider $f(0.5) \in S^1$. Now take the set $S^1 \setminus \{f(0.5)\}$ is a connected set. (If a set in $\mathbb{R}^2$ is path connected $\iff$ connected)

As f^{-1} is ~~connected~~ continuous, $f^{-1}(S^1 \setminus \{f(0.5)\})$ is connected

$= [0, 0.5) \cup (0.5, 1]$

$\hookrightarrow$ disconnected set $\to\leftarrow$

Lec-12

Wednesday

Connectedness

A connected set is one that is "in one piece".

Example :- $(\mathbb{R}, d)$ is usual metric

① $\mathbb{R}$ → connected

② $[0,1] \cup [0,1), (0,1)$ → connected sets.

③ $[0,1] \cup [1,2]$ → connected $\cap \neq \phi$

④ $(0,1) \cup [1,2]$ → not connected $\cap = \phi$

⑤ $[0,1) \cup [1,2]$ → connected, but $\cap = \phi$

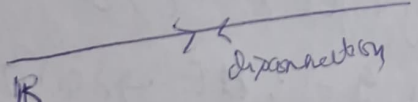
Definition :- Let (M,d) be a Metric space we say M is disconnected if M can be written as disjoint union of two open sets. In other words,

$M = A \cup B$, where

$A \cap B = \phi$ + A and B are open.

And if we cannot find such disconnection in M, then we call M is connected.

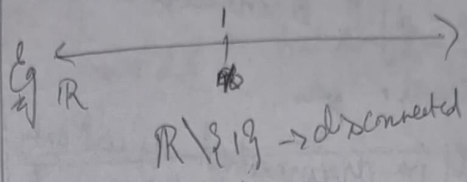
Remark: If M is disconnected and A, B open s.t $A \cap B = \phi$ and $M = A \cup B$; then A and B are called disconnection of M.



Remark :- If $E \subseteq M$, to say E is disconnected, we need to consider the disconnection on (E,d)

Remark :- The disconnection sets are both open and closed in (M,d)

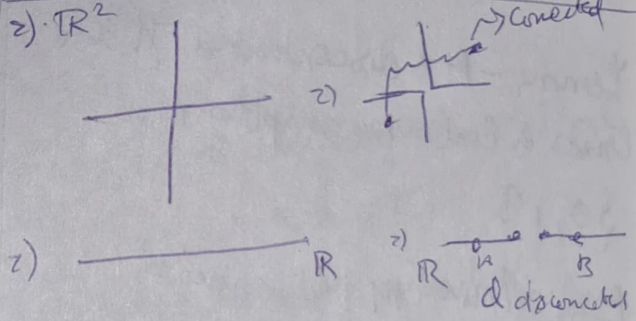
Theorem :- Let (M,d) be a metric space M is connected if there is no nontrivial (neither M nor ϕ) clopen (closed + open) subset in M.



$\mathbb{R} \setminus \{1\}$ → disconnected

Result :- Let $A \subset M$; If U and V are disjoint open sets in A. Then there are two disjoint open sets in M given by L and K such that

$U = L \cap A$ and $V = K \cap A$



Properties of Connectedness

New Defn :-

$A \subseteq M$ is disconnected if there are two disjoint open sets. U and V in (M,d) s.t

$A \cap U \neq \phi$

$A \cap V \neq \phi$ and $A \subset U \cup V$

Exam :- $[0,1] \cup (1,2]$ → disconnected

Sol, there are two disjoint open sets in

$\mathbb{R}$. $(-\infty, 1) \& (1, \infty)$

Theorem:- A subset E of $\mathbb{R}$, containing more than one point is connected iff for any $x, y \in E$, say $x < y$. we have $[x, y] \subset E$.

Furthermore: if E is connected then E is an interval.

~~Proof:- M is disconnected if and only if f a continuous map onto M onto $\{0, 1\}$~~

proof:→ Later

Lemma:- M is disconnected if there exists a continuous map f onto M onto $\{0, 1\}$.

Proof:- Given M is disconnected

$M = M_1 \cup M_2$, where M_1 and M_2 are open and $M_1 \cap M_2 = \phi$.

Define the map

$$f(x) = \begin{cases} 0 & x \in M_1; \\ 1 & x \in M_2. \end{cases}$$

↳ Continuous fun and onto map

Conversely - Assume there is a $f: M \to \{0, 1\}$ Continuous and onto

$A = f^{-1}(\{0\}) \longrightarrow$ clopen set

$B = f^{-1}\{1\} \to$ clopen set

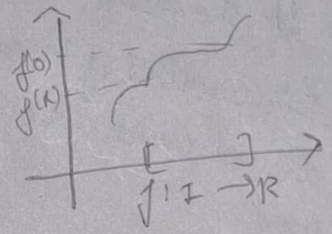
- Symb, $A \cap B = \phi$

$+ M = A \cup B$

Hence M is disconnected (proved)

Remark :- (M, d) is connected iff any continuous map from M into discrete space is constant.

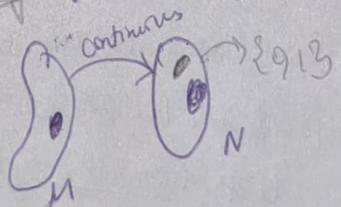
Note → Discrete space means any subset is open as well as closed



Theorem:- Let $f: (M, d) \to (N, \rho)$ be a continuous map.

If $E \subset M$ is connected then $f(E)$ is connected.

proof:- Assume $f(E)$ is not connected



Then there is a continuous onto map $g: f(E) \to \{0, 1\}$

Consider $g \circ f: E \to \{0, 1\}$

↳ continuous fun (Check)!

and $g \circ f$ is also onto.

By previous theorem

E is disconnected $\Rightarrow \Leftarrow$
Contradiction.